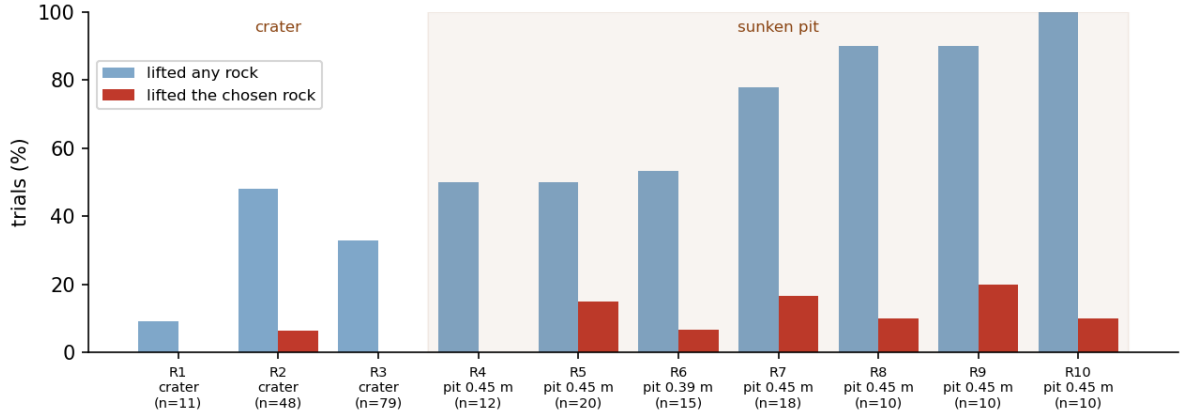
Excavator rock pick-up: trials and failures

Isaac Sim 5.1, four-joint excavator. 26 October 2026. Every number here comes from `docs/pick_rock/trials.json` (237 headless trials).

Summary

Goal: a motion that picks up a rock from the pile and holds it in the raised bucket (rock centre above 10 cm after a 2 s hold). Strict version: pick a *chosen* rock.
Result: the final motion, an edge-first drag into a **sunken pit**, lifts and holds rocks in **10 of 10** trials (2 to 23 rocks per scoop). It lifts the specific rock it aimed at in only **1 of 10**: aiming at one rock is still unreliable.
Correction (same day): the first version of this report gave 9/10 and 2/10 for a motion that turned the bucket the wrong way round, through the stick, during the approach, and tilted it past its range while lifting. The URDF joints have no limits and the simulator doesn't collide neighbouring links, so nothing stopped it. The motion now stays inside collision-free joint ranges measured from the meshes; the numbers above are for the corrected motion (round R10).
What made the difference: (1) keeping every vertex of the bucket out of the dirt during the curl (otherwise it jammed); (2) sinking the rocks into a pit below ground, so the lip passes under them instead of bulldozing them across flat ground; (3) tilting the bucket back while lifting so the load doesn't spill over the lip; (4) refusing the IK solution that swings the cab round 180 degrees.



Success rate per round. On the crater pile no motion picked up a rock reliably; in the pit nearly every final-motion scoop holds rocks, but the aimed-at rock is caught only sometimes. R6 is the pit moved closer to the machine.

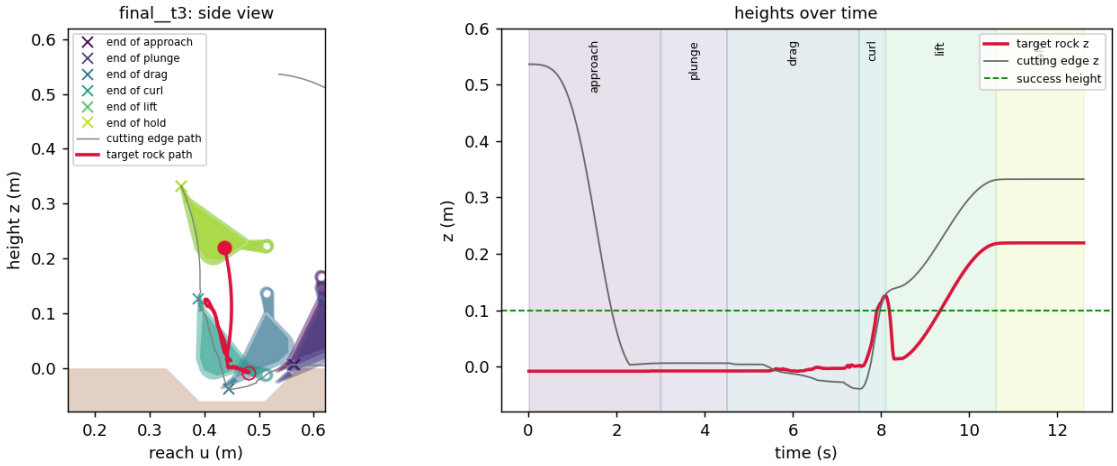
The final motion

Planned entirely from kinematics (open loop) by `scripts/pick_motion.py` as one array of joint targets, 60 per second, about 10.6 s long, every target inside the joint ranges below. Pitch is the bucket angle (`arm - pole - bucket`): 2.5 means the cutting edge points down and back at the machine, 0.5 means the bucket opens straight up.

Phase	Time	What the bucket does
approach	3.0 s	Joint move to above the entry point, 8 cm beyond the rock (less if out of reach), pitch 2.5
plunge	1.5 s	Edge straight down to 3 cm below the rock's centre (never closer than 8 mm to the dirt)
drag	3.0 s	Edge pulled toward the machine to 5 cm short of the rock, following the pit profile; whole bucket mesh kept 5 mm clear of the dirt
curl	0.6 s	Rotate to pitch 0.5 about the edge while it moves 4 cm further in, raised only as much as the clearance needs
lift	2.5 s	Boom up 26 degrees (edge rises about 20 cm) while tilting back toward pitch -0.2, stopping 3 degrees before the bucket meets the stick (in practice pitch 0.5-0.8)

Joint	Collision-free range	What it hits outside
boom (arm_joint)	-50 to 110 deg	cab
stick (pole_joint)	-5 to 130 deg	boom (faces touch, at most 0.4 mm, inside the range)

Joint	Collision-free range	What it hits outside
bucket (bucket_joint)	-8 to 278 deg	stick, between -80 and -10 deg: a pose at -160 deg has to be reached as +200 deg



Successful corrected trial (target rank 3, 6 rocks in the bucket). Left: bucket at the end of each phase (from the commanded angles) over the pit profile, and the target rock's path (red). Right: target rock height and cutting-edge height over time.

Rounds of trials

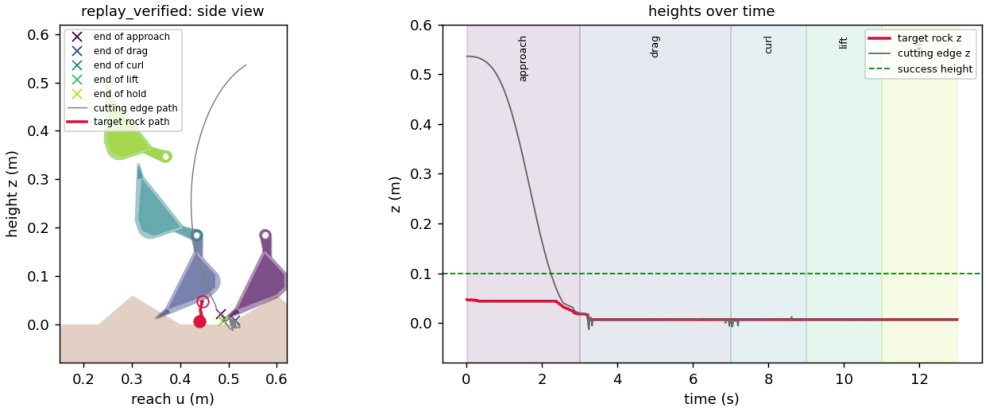
	Scene	Tried	n	Chosen rock	Any rock	Most rocks	Outcome
R1	crater	Old joint-space dig cycle, straight-down grab, edge-first drags (v1-v7)	11	0	1	1	Bucket jammed in the crater wall and later the heel in the ground; edge too shallow, rock bulldozed
R2	crater	Grid: entry/exit distance, depth, lip advance, curl time (48 combos)	48	3	23	3	3 isolated successes (fast 1 s curl, edge at ground level); neighbours failed
R3	crater	Finer grid over 5 target rocks each (partial run, stopped)	79	0	26	2	No target picks: on flat ground loose rocks are pushed ahead of the lip
R4	pit 0.45 m	Sunken pit, drag follows pit profile (v1-v4)	12	0	6	26	Scoops 12-27 rocks, but target rides on top and spills; far entries swing round
R5	pit 0.45 m	Forward-only IK, tilt bucket back while lifting (v4-v7)	20	3	10	28	Tilt-back keeps the target in; far-side targets out of reach
R6	pit 0.39 m	Pit moved closer to fix reach	15	1	8	11	Worse: the bucket could not push down into the packed, walled-in rocks
R7	pit 0.45 m	Back to 0.45 m; entry point slides in until reachable; 6 targets	18	3	14	27	v6 (carry pitch -0.2) best: 3/6 targets
R8	pit 0.45 m	Final motion (= v6) on 10 targets	10	1	9	26	1/10 target; one target unplannable
R9	pit 0.45 m	Final motion, targets chosen by full plannability	10	2	9	24	2/10 target, 9/10 any rock, but the bucket passed through the stick (see Correction)
R10	pit 0.45 m	Corrected: every joint kept inside its collision-free range; lift with the boom	10	1	10	23	1/10 target, 10/10 any rock

Not counted: 4 reruns while debugging the camera (below). Repeating the same trial gives the same result within one run, but results shift a little with run order (the pile resets to within a few millimetres), so single successes are not proof; that is why later rounds use 5 to 10 target rocks.

Failures, in order

1. Old dig cycle jams in the crater (R1)

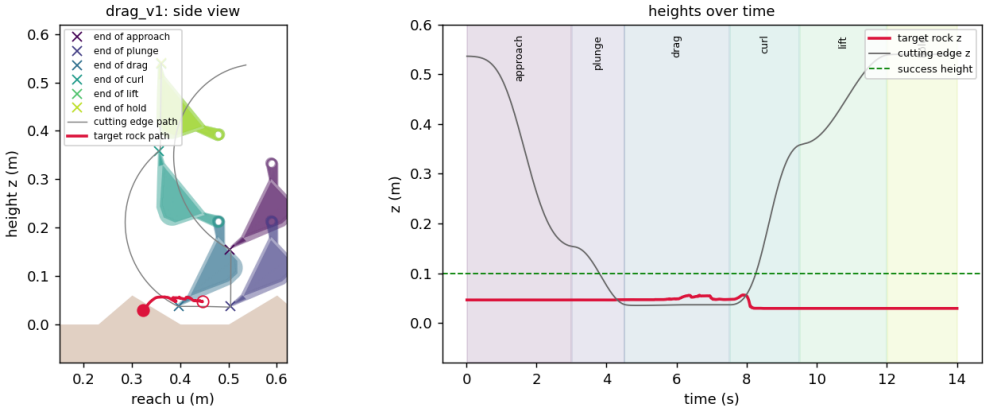
The joint-space cycle verified on an earlier pile drove the bucket into the crater's sloped inner wall. The pole stopped at 0.03 rad against a commanded 1.19 and nothing was lifted.



replay_verified: the cutting edge (grey) never follows the plan; the robot is stuck.

2. Edge too shallow: the rock is pushed, not scooped (R1)

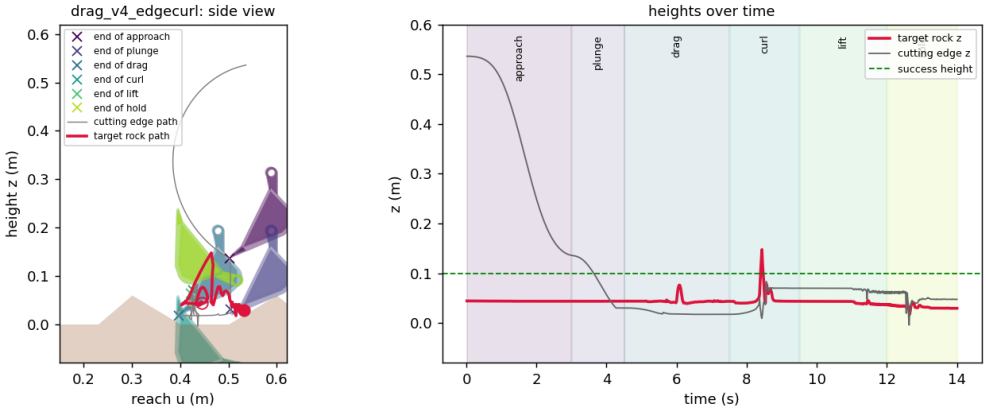
Aiming the edge 8 mm below the rock's centre only shoved it along. The curl, a pure bucket-joint rotation, then swung the lip up through it and knocked it toward the machine.



drag_v1: the rock (red) moves with the edge, then drops off during the curl.

3. Curl drives the heel into the ground (R1)

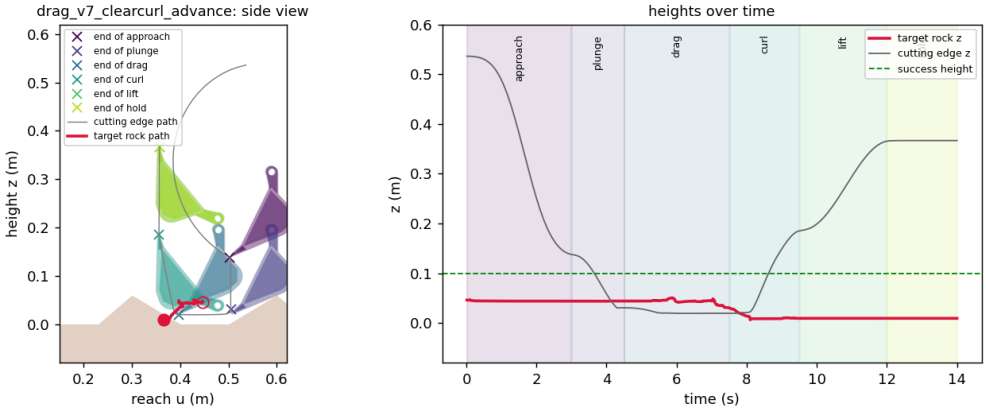
Rotating the curl about the cutting edge got the rock up to 15 cm, but the bucket's heel went about 2 cm below the ground and the arm jammed: commanded lift, no motion. A heel check against flat ground alone was not enough, because the bucket's back then went into the crater wall (joint errors up to 0.8 rad). The fix was to check every bucket mesh vertex against the dirt surface.



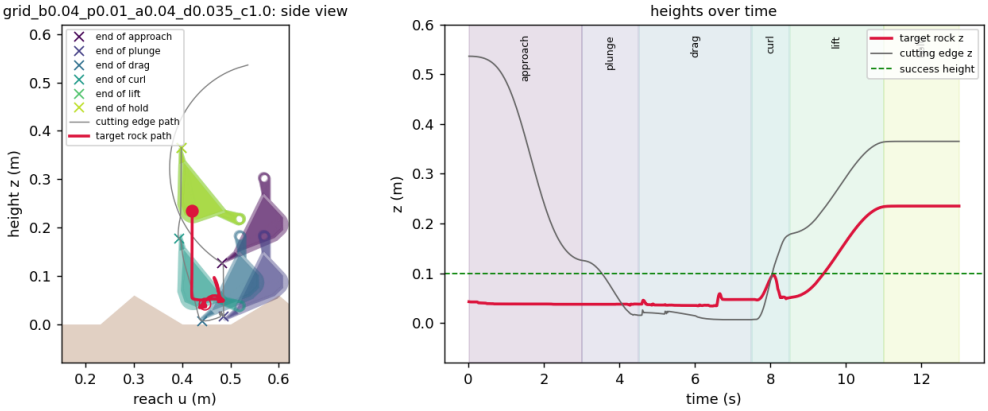
drag_v4_edgecurl: the edge (grey) stays at 7 cm through the lift; the plan raised it to 25 cm.

4. No more jams, but loose rocks are bulldozed (R1-R3)

With clearance enforced, the motion completes, but on the crater the rock sits at the lip and gets pushed ahead of it until it drops off the front of the heap. The 48-combination grid found 3 successes, all isolated (their neighbours failed). The finer grid caught no target rock in 79 trials.



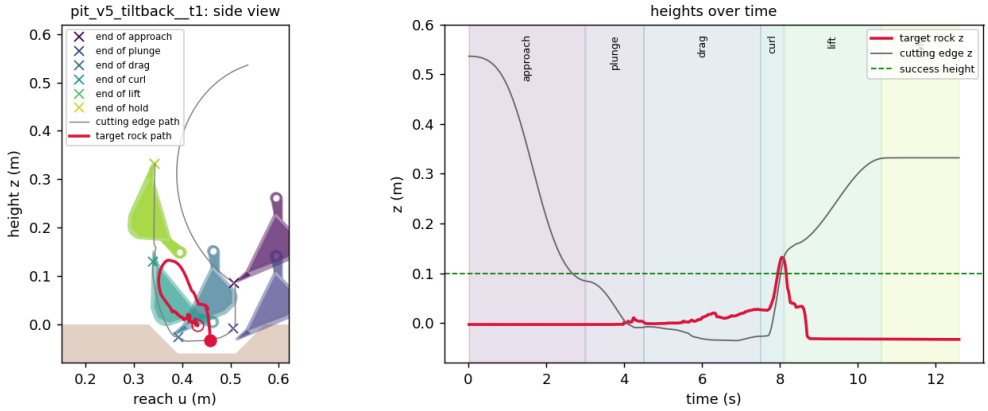
drag_v7: the motion completes; the rock is pushed off the front of the heap before the curl.



One of the 3 crater successes: the rock rode the lip into the bucket during a fast curl.

5. Sunken pit: big scoops, but the target spills (R4-R5)

Rocks poured into a 6 cm deep pit stay held by its walls, so the lip passes under them: scoops of 12 to 28 rocks. (The ground plane is solid, so the pit is a static dirt slab at ground level with the hole cut out, and the real ground plane is lowered under it.) The aimed-at rock often sat on top of the heaped load and slid over the lip while lifting. Lifting with the bucket tilted back (pitch -0.2) fixed most of that.

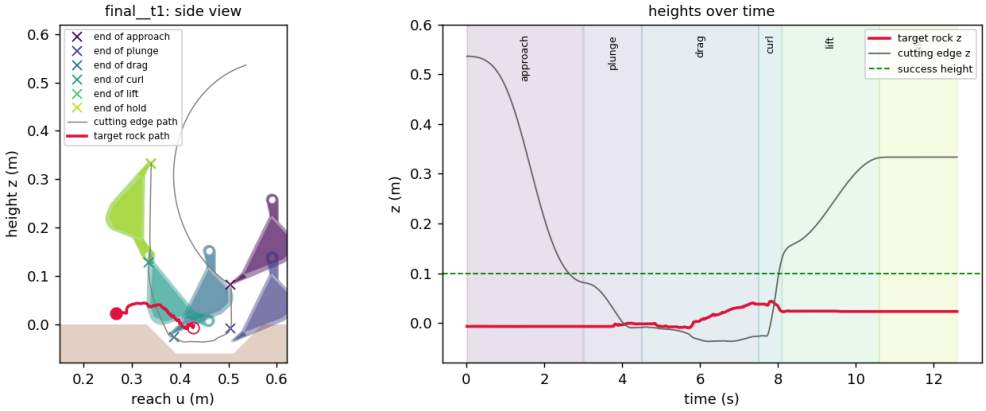


pit_v5 (tilt back to 0.1), 27 rocks lifted: the target reached 13 cm, then slid out. Pitch -0.2 holds better.

6. Reach: wrong IK branch, then a pit that is too close (R5-R7)

Entry points beyond the rock on the far side of the pit were out of forward reach (about 0.55 m at pit depth), and IK quietly picked the other solution, swinging the cab round 180 degrees to reach back over it. That branch is now refused. Moving the pit to 0.39 m fixed reach but made digging worse. The commanded edge went to -29 mm, but the real one stopped at +5 mm, resting on rocks that the walls held in place. The pit went back to 0.45 m, and the entry point now slides toward the rock until it is reachable.

7. Final motion: rocks yes, the chosen rock only sometimes (R8-R9)



Final motion, target rank 1: 24 rocks lifted, but not the target, which stayed beside the scoop.

8. The bucket went through the stick (found after the first report)

Watching the motion live showed the bucket turning nearly a full circle and passing through the stick. Cause: the joints are continuous and IK wrapped angles to the 'nearest' equivalent, so reaching the dig pose (bucket about -160 deg) turned the bucket through -10 to -80 deg, where it overlaps the stick. Fix: rotating each link mesh against its neighbour gave collision-free ranges, now in `kinematics.JOINT_LIMITS`. IK picks the in-range equivalent (the bucket turns the other way, +200 deg), pitch blends take the short way round, and the lift raises the boom instead of moving the edge straight up, which ran out of range. A whole-plan mesh check now finds no link overlap. Keyboard clamps and sweep ranges in the control scripts were changed to the same ranges (the old sweep also turned the bucket into the stick). Remaining: joints lag up to 0.5 rad while the bucket makes its long 207 degree turn in the 3 s approach.

Side issue: no camera pictures

Headless renders from an Isaac camera sensor came out empty: only sky and a dark plane. This happened even with lights added, the camera aimed by look-at before or after reset, Fabric off, and rendering every step. The same code renders correctly in a minimal scene, so the cause is something about this scene that I did not find. The figures here are drawn from logged data instead: the bucket is the real mesh, posed from the commanded joint angles.

Next steps

• Aim the cut at the rock's side, not its centre, so the target lands in the middle of the bucket rather than on the heaped top; or make the scoop shallower so fewer rocks compete for room.
• Close the loop: re-plan the drag from the target's live position (it moves 2-5 cm while the edge approaches).
• Fix the joint drive tracking (about 9 mm mean lag at kp=1e4) or slow the curl, so the real edge follows the plan.
• Put the joint ranges into the URDF (revolute with limits) and re-import, so the simulator itself stops a link at its range.
• Try the dump to the bin (+90 degree swing) now that the bucket holds a load.

How to reproduce

`~/isaacsim/python.sh scripts/pick_rock.py final --targets=10` (headless, Isaac Sim closed). Then `/usr/bin/python3 scripts/plot_pick.py` for the plots. In `run_live.py`, press **P** to run the same motion live (restart needed: the pit moves the ground plane).